Homework — 1.5.1 Computing Related Legislation — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] 1 mark: Computer Misuse Act 1990, **Offence 1** (unauthorised access to computer material). 1 mark for justification: the nurse had no authorisation to view *those* records / access for a non-work purpose is unauthorised even though their login may exist; nothing was modified and no further offence was intended, so it is Offence 1 only. *(Accept a credited note that the DPA is also engaged by the trust if records are misused, but the offence the individual commits is CMA Offence 1.)*

2. [3] 1 mark each: (a) Data subject = the living individual the personal data is about — e.g. the app user whose heart rate is recorded. (b) Data controller = the organisation that decides why/how the data is processed — e.g. the fitness-app company. (c) Data processor = a party that processes the data on the controller's behalf — e.g. a cloud-hosting firm storing the heart-rate records. Example must fit the scenario for each mark.

3. [2] 1 mark: yes — the developer still **owns the copyright** (copyright arises automatically on creation and is retained by the author). 1 mark: the open-source **licence** is the mechanism by which the copyright holder *grants others permission* to copy/modify the work; without that licence such copying would infringe their copyright. *(Reasoning that "open source ≠ no copyright" is the key idea.)*

4. (a) [2] Up to 2 from: interception is a serious intrusion on privacy / a powerful state power **[1]**; restricting who may authorise it provides accountability/oversight, ensures it is proportionate and used only for legitimate purposes (national security, serious crime), and helps **prevent abuse** of the power **[1]**.

(b) [1] Any one: require ISPs/communications providers to give access to communications data; demand that a person hand over encryption keys; lawfully intercept/monitor communications for an authorised purpose.

5. [2] 1 mark: Computer Misuse Act 1990, **Offence 3** (unauthorised modification of computer material). 1 mark: deleting/altering the database is an unauthorised *modification*; the Act is about *authorisation* and the technician was no longer authorised to act on the system (and never authorised to destroy data) — it does not require a password to be stolen, only that the modification was unauthorised. *(Accept that intent/scheduling could also engage Offence 2.)*

6. [2] 1 mark: **Copyright, Designs & Patents Act 1988** — using other people's photographs without a licence/permission infringes the rights-holders' copyright. 1 mark: **Data Protection Act 2018 / GDPR** — keeping users' names and emails for ten years "just in case" breaches the *storage limitation* (keep no longer than necessary) and/or *data minimisation* principle. Must link each action to the correct law.

Part B (6 marks)

7. [2] 1 mark + 1 mark for a clear difference, e.g. symmetric encryption uses the **same single key** to encrypt and decrypt (which must be shared securely) **[1]**; asymmetric uses a **key pair** — a public key to encrypt and a different private key to decrypt, so no secret key need be shared **[1]**. *(Accept: symmetric is faster but has the key-distribution problem; asymmetric is slower but solves key exchange.)*

8. [2] 1 mark: a stack is **LIFO** — the last item pushed on is the first popped off (removed from the top). 1 mark: a queue is **FIFO** — the first item enqueued is the first dequeued (removed from the front).

9. [2] 1 mark + 1 mark, any one developed point: high-level code is closer to English / easier and faster to write, read and maintain **[1]**; it is portable across different processors whereas assembly is tied to one architecture / fewer instructions needed / built-in features (data structures, libraries) handle low-level detail for you **[1]**.

Total: 14 + 6 = 20 marks